Project Title :

Health Care and Medicine Guide

Description :

The Health Care and Medicine Guide is a project that aims to provide users with valuable health information and tools to improve their lifestyle and well-being. The website includes multiple sections such as Health Articles, Health Score Checker, and a Medication Guide. Users can read health tips, calculate their health score based on lifestyle factors and medical history, and look up detailed information about best lifestyle. The website will be built using HTML, CSS, JavaScript, and SQL for the database. It includes both frontend and backend functionalities, ensuring a complete full-stack web application. Additionally, an ***Admin Panel*** *will be implemented to allow administrators to publish and manage blog articles by category.*

Objective of the project :

The primary objective of this project is to create an interactive and user-friendly website.

- *Educates users about health and well-being through informative articles.*
- *Provides a Health Score Checker system to analyse users health status.*
- *Offers a Medication Guide for users to search for medicines and understand their usage, side effects, and precautions.*
- *Allows administrators to manage and publish blog articles through an Admin Panel.*
- *Stores user data in a SQL database for future reference and analysis.*
- *Promotes a healthier lifestyle through personalized recommendations.*

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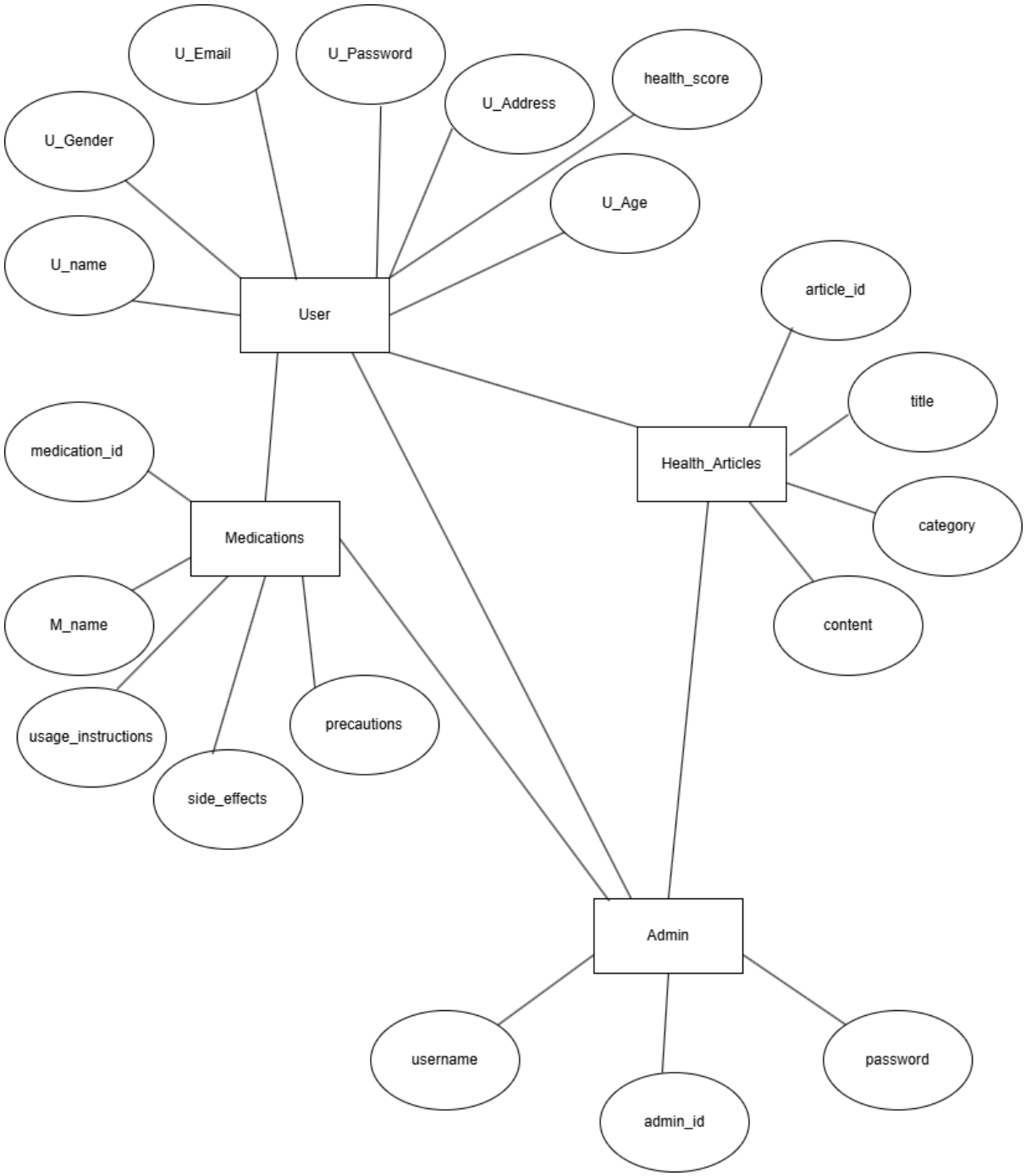
Entities of project :

- ***User*** *: The end-user accessing the website.*
- ***Health Articles*** *: A collection of categorized health tips and medical articles.*
- ***Admin Panel*** *: A secure section for administrators to publish and manage articles by category.*
- ***Health Score Checker*** *: A form-based system to calculate a user's health score based on inputs like age, weight, height, lifestyle habits, and medical history.*
- ***Medication Guide*** *: A searchable database of medications with details such as usage, side effects, and precautions.*

ERD Diagram:



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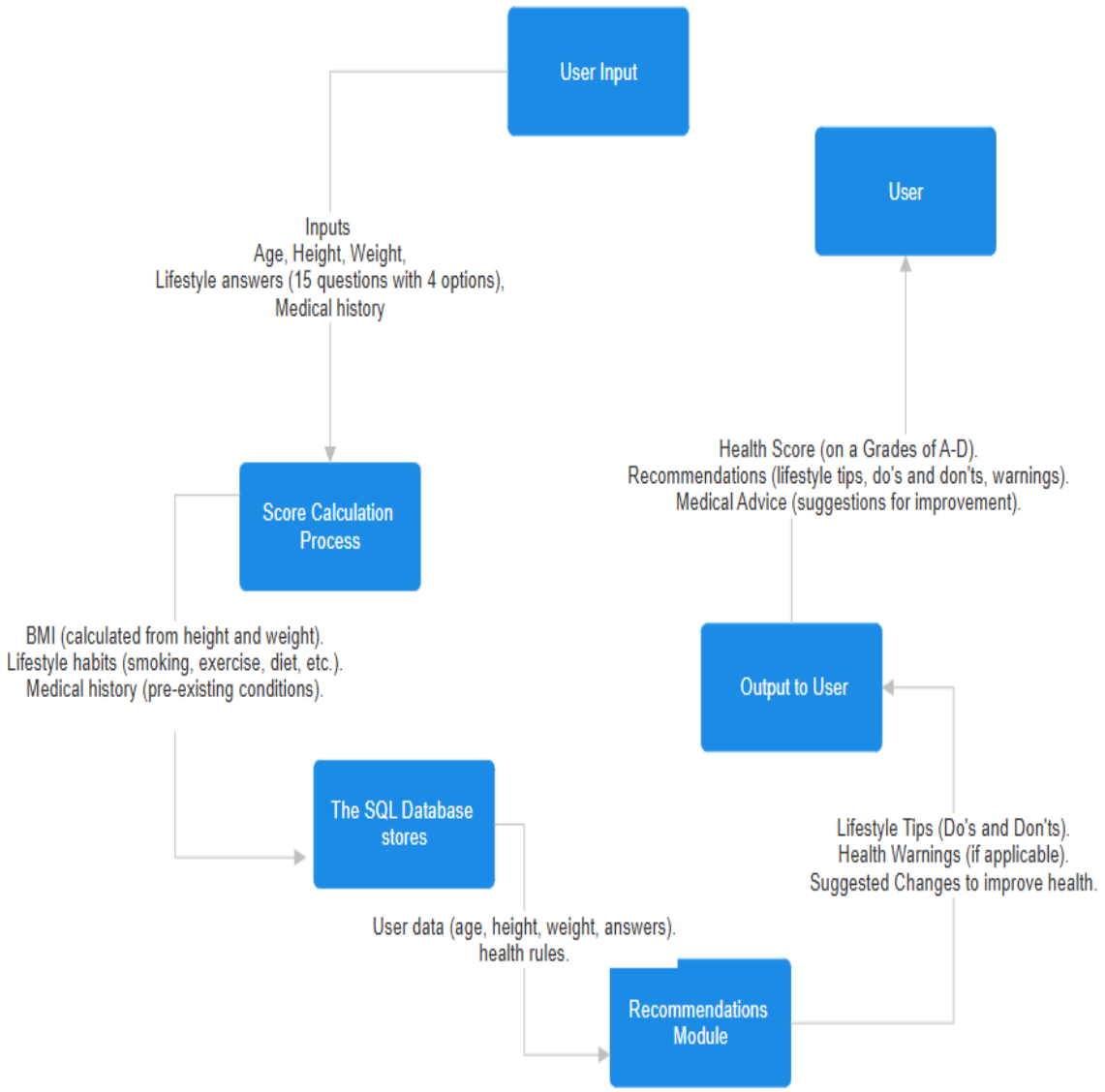
DFD Diagram:



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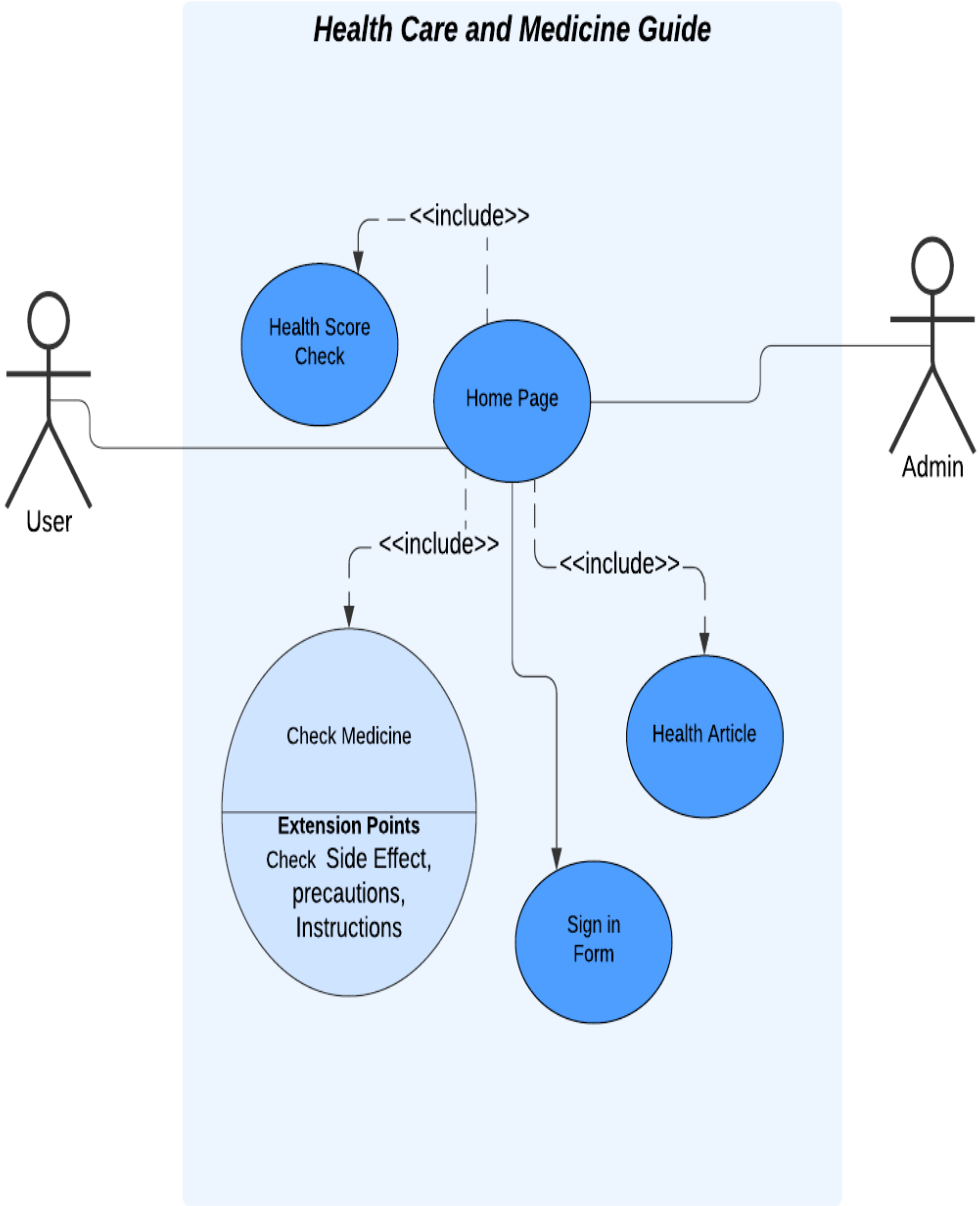
Use Case Diagram:



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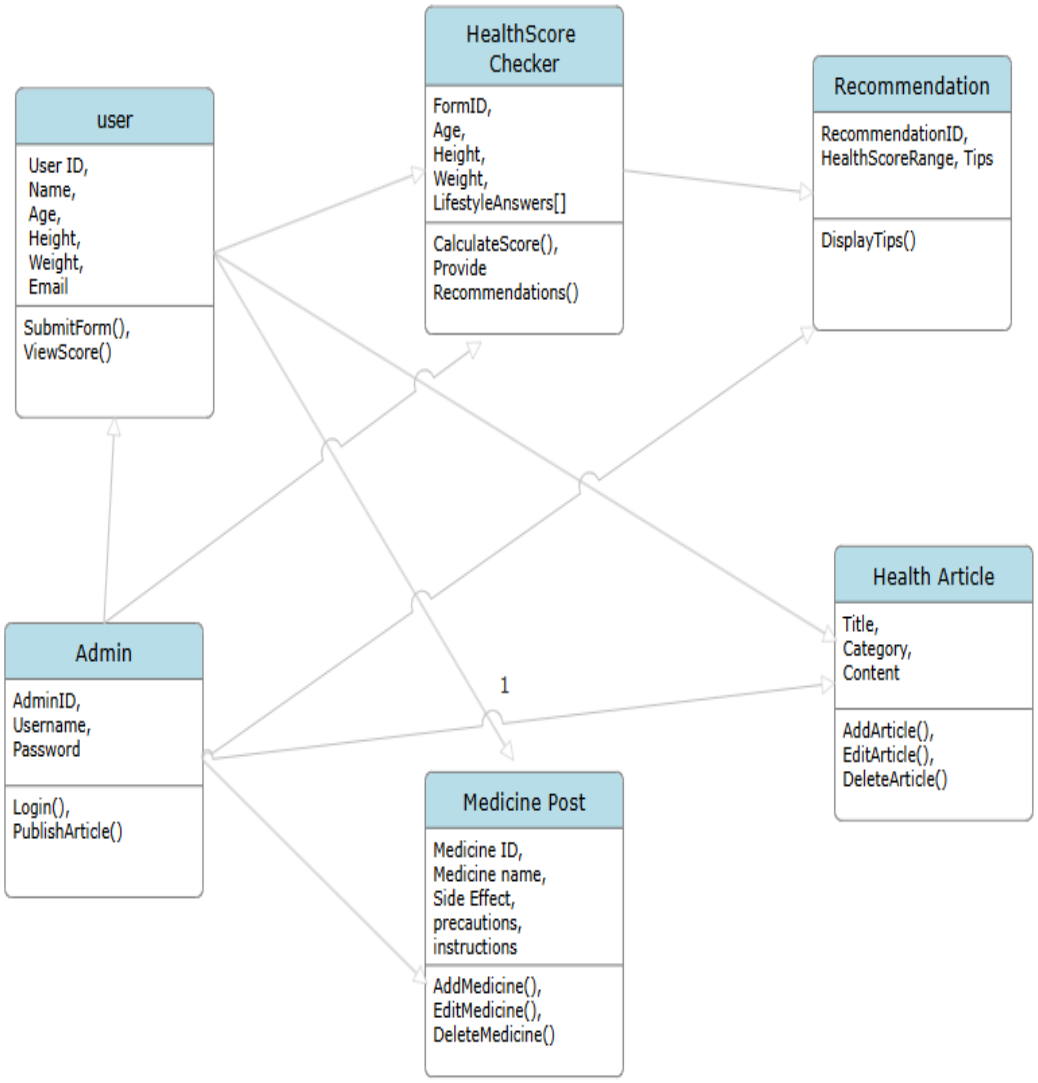
Class Diagram:



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Frontend and Backend Technologies :

Frontend :

- *HTML: For structuring the webpages.*
- *CSS: For styling the pages.*
- *JavaScript: For interactivity and dynamic functionality.*

Backend :

- *SQL: For storing and managing data in the database.*
- *Admin Panel: A secure backend section for managing health articles.*

Tools and IDE :

- *Visual Studio Code: For coding the project.*
- *Live Server: For testing and viewing the website in the browser.*

Test Plan :

- ***Unit Testing:*** *Test individual components like forms, search functionality, etc.*
- ***Integration Testing:*** *Ensure all components work together seamlessly.*
- ***System testing****:*
- **Verification of System Requirements:** *Ensures that the software meets the specified requirements and functions correctly.*
- ***Validation of System Processes:*** *Confirms that the system supports the business processes as intended.*
- ***Database Testing:*** *Verify that the SQL database is storing, retrieving, and updating data correctly.*
- ***Admin Panel Testing:*** *Ensure that the admin panel works securely and allows proper management of articles.*
- ***User Acceptance Testing:*** *Ensure the website meets the user's requirements and expectations.*

Execution of Development :

1. ***Requirement Analysis:*** *Gather and analyse the project requirements.*
2. ***Design:*** *Create ERD, DFD, and UML diagrams.*
3. ***Development:***
    - ***Frontend Development:*** *Build HTML, CSS, and JavaScript components.*
    - ***Backend Development:*** *Set up the SQL database and write queries to interact with the frontend.*
    - ***Admin Panel Development:*** *Create a secure admin interface for publishing and managing articles.*
4. ***Testing:*** *Conduct testing based on the test plan.*
5. ***Deployment:*** *Deploy the project on a local or online server.*

Development Process :

Phase 1: Setup and Basic Structure:

- *Design the homepage, Health Articles page, Health Score Checker form, and Medication Guide page and Sign In Page form.*
- *Use CSS to style the pages and make them responsive.*

Phase 3: Backend Development:

- *Set up the SQL database.*
- *Create tables for storing user data, articles, and medications.*
- *Write SQL queries to insert, update, and retrieve data.*
- *Develop the Admin Panel for managing articles.*

Phase 4: JavaScript Logic:

- *Implement form validation for the Health Score Checker.*
- *Add logic to calculate the health score based on user inputs.*
- *Implement the search functionality for the Medication Guide.*

- *Add interactive features in the Admin Panel (e.g., CRUD operations).*
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Phase 5: Testing and Debugging:

- *Perform unit and integration testing.*
- *Fix any bugs or issues found during testing.*
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Phase 6: Deployment:

Deploy the project on a local server using Live Server or deploy it online using GitHub Pages or any hosting platform.